

CMPE2600 ICA01 Simple Program

In this exercise you are going to create a program with 4 inputs and 2 outputs. SW1 and SW2 should be assigned to Inputs 11 and 12, which are normally open (N.O.) Momentary. SW3 and SW4 should be assigned to inputs 8 and 9 which are normally closed (N.C.).



Your Tags should look something like this:

Name	Alias For	Base Tag	Data Type	Description	External Access	Constant	Style
Local:1:C			AB:Embedded_Di...		Read/Write	☐	
Local:1:I			AB:Embedded_Di...		Read/Write	☐	
Local:1:O			AB:Embedded_Di...		Read/Write	☐	
Local:2:C			AB:Embedded_An...		Read/Write	☐	
Local:2:I			AB:Embedded_An...		Read/Write	☐	
Local:2:O			AB:Embedded_An...		Read/Write	☐	
Local:3:C			AB:Embedded_HS...		Read/Write	☐	
Local:3:I			AB:Embedded_HS...		Read/Write	☐	
Local:3:O			AB:Embedded_HS...		Read/Write	☐	
SW1	Local:1:I.Data.11	Local:1:I.Data.11	BOOL		Read/Write	☐	Decimal
SW2	Local:1:I.Data.12	Local:1:I.Data.12	BOOL		Read/Write	☐	Decimal
SW3	Local:1:I.Data.8	Local:1:I.Data.8	BOOL		Read/Write	☐	Decimal
SW4	Local:1:I.Data.9	Local:1:I.Data.9	BOOL		Read/Write	☐	Decimal
Light_1	Local:1:O.Data.1	Local:1:O.Data.1	BOOL		Read/Write	☐	Decimal
Light_2	Local:1:O.Data.2	Local:1:O.Data.2	BOOL		Read/Write	☐	Decimal
						☐	

Prior to downloading the program, answer the following questions.

What must the states of the input devices SW1-SW4 be in order to turn on LIGHT1? (i.e. Pressed or Not Pressed?)

SW1 must be pressed.

SW2 must be pressed.

SW3 must be pressed.

SW4 must be pressed.

What must the states of the input devices SW1-SW3 be in order to turn on LIGHT2? (i.e. Pressed or Not Pressed?)

SW1 must be pressed.

SW2 must be not pressed.

SW3 must be not pressed.

Demonstrate your working program on a PLC to your instructor for a check-off. Instructors Initials: _____